Pstat131 hw1

Elliott Liu

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Question 1

Define supervised and unsupervised learning. What are the difference(s) between them? Supervised learning uses data with a known response variable Y, the target, and predictors X, and aims to find out the relationship $Y = f(X) + \epsilon$ for prediction, estimation, model selection, and inference.

Unsupervised learning doesn't have response variable Y and only have the predictors X.

The differences between them are that the goal of supervised learning is to predict or explain the response, while the goal of unsupervised learning is the discover patterns or structure in the data. Also, another obvious difference is that whether the labeled responses are available.

Question 2

Explain the difference between a regression model and a classification model, specifically in the context of machine learning. A regression model is used when the response variable Y is quantitative, such as predicting price, blood pressure, speed, and so on, while a classification model is used when the response variable Y is categorical or qualitative, such as predicting whether a nodule is benign or malignant, or if a phone call is spam or not, and so on.

Question 3

Name two commonly used metrics for regression ML problems. Name two commonly used metrics for classification ML problems. Regression models are usually evaluated by metrics such as $RMSE$ and R^2; however, classification models are evaluated using Accuracy and Area under ROC curve.

Question 4

As discussed, statistical models can be used for different purposes. These purposes can generally be classified into the following three categories. Provide a brief description of each. Descriptive models are used to show or visualize trends in the data, such as using time series plots or scatter plots.

Inferential models are used to test theories (possibly causal claims) and state the relationship between outcomes and predictors, such as linear, causal, and so on. It's also used to identify significant features.

Predictive models are aim to predict Y ($E[Y|X]$) with minimum reducible error, without focusing on hypothesis test, and predict what combination of features fits best.

Question 5

Predictive models are frequently used in machine learning, and they can usually be described as either mechanistic or empirically-driven. Answer the following questions.

Define mechanistic. Define empirically-driven. How do these model types differ? How are they similar? Mechanistic models assume a fixed parametric form of f and are often based on knowledge or theories, while empirically-driven models, which normally learn structure from the data itself, make little or no assumptions, and they are more flexible by default and non parametric. Both models are used to predict the relationship between predictors and outcome variables and may have a problem of overfitting.

In general, is a mechanistic or empirically-driven model easier to understand? Explain your choice. Mechanistic models are less flexible but easier to interpret; however, empirically-driven models are more flexible and often need a larger number of observations. Therefore, in general, mechanistic models are easier to understand because their structure and parameters have clear interpretations.

Describe how the bias-variance tradeoff is related to the use of mechanistic or empirically-driven models. Mechanistic models usually have higher bias and smaller variance. In contrast, empirically-driven models tend to have smaller bias but higher variance because it's more flexible.

Question 6

A political candidate's campaign has collected some detailed voter history data from their constituents. The campaign is interested in two questions:

Given a voter's profile/data, how likely is it that they will vote in favor of the candidate? This is predictive, because it asks to predict an outcome which is the probability that a voter will support the candidate given their profile.

How would a voter's likelihood of support for the candidate change if they had personal contact with the candidate? This is inferential. Instead of simply predict based on the information or the data, it foucusing on understanding a relationship of a specific factor (the personal contact).

Exploratory Data Analysis

Exercise 1

We are interested in highway miles per gallon, or the hwy variable. Create a histogram of this variable. Describe what you see/learn.

```r
library(tidyverse)
```

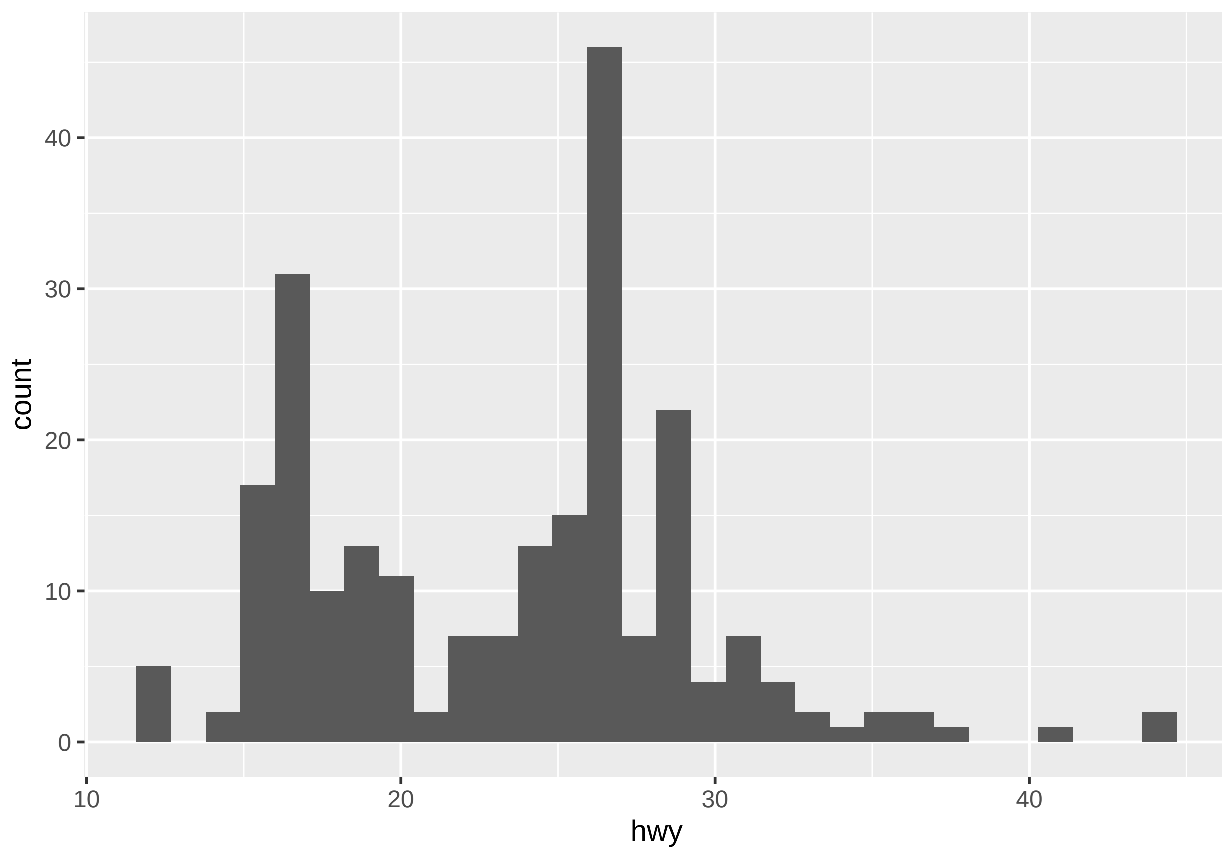
```
## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
mpg
```

```
## # A tibble: 234 x 11
##    manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
##    <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
##  1 audi         a4           1.8  1999     4 auto~ f        18    29 p     comp~
##  2 audi         a4           1.8  1999     4 manu~ f        21    29 p     comp~
##  3 audi         a4           2    2008     4 manu~ f        20    31 p     comp~
##  4 audi         a4           2    2008     4 auto~ f        21    30 p     comp~
##  5 audi         a4           2.8  1999     6 auto~ f        16    26 p     comp~
##  6 audi         a4           2.8  1999     6 manu~ f        18    26 p     comp~
##  7 audi         a4           3.1  2008     6 auto~ f        18    27 p     comp~
##  8 audi         a4 quattro   1.8  1999     4 manu~ 4        18    26 p     comp~
##  9 audi         a4 quattro   1.8  1999     4 auto~ 4        16    25 p     comp~
## 10 audi         a4 quattro   2    2008     4 manu~ 4        20    28 p     comp~
## # i 224 more rows
```

```r
library(ggplot2)
mpg %>% ggplot(aes(x = hwy)) +
  geom_histogram()
```

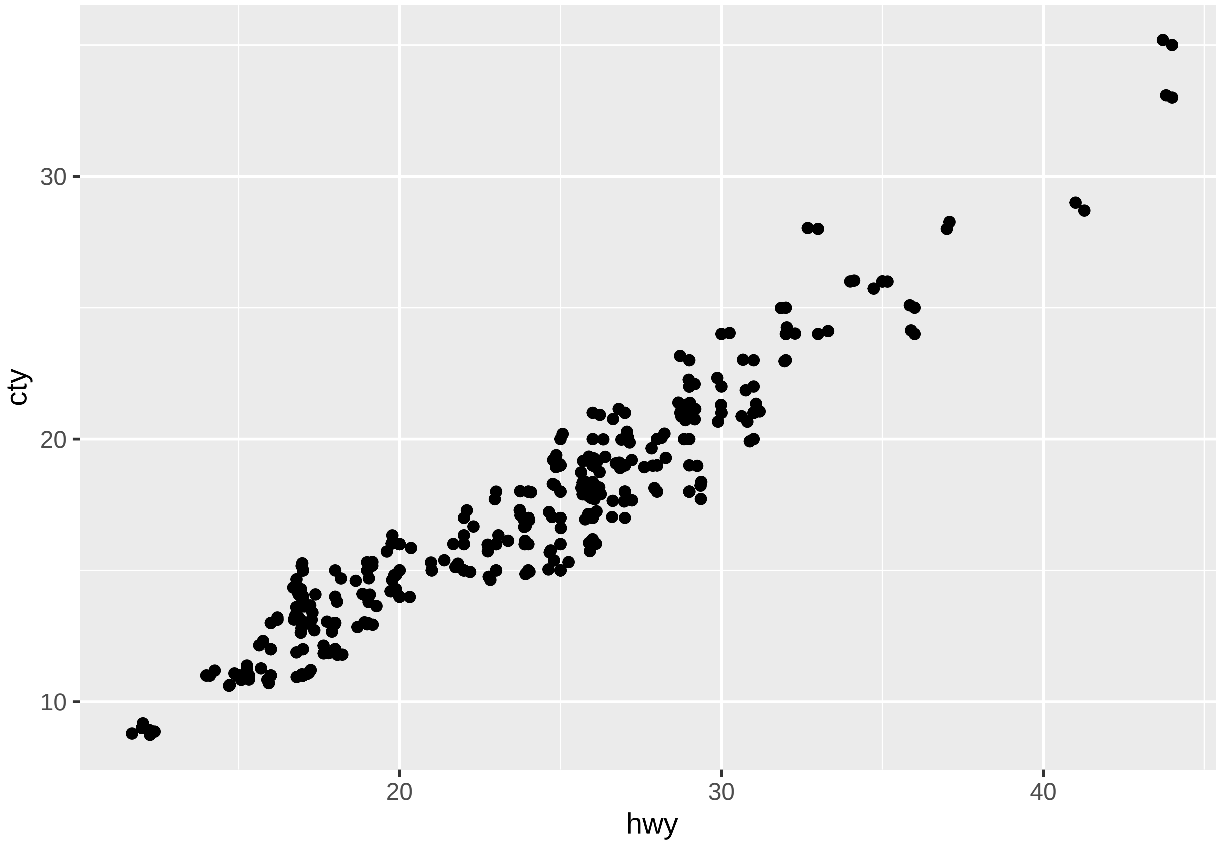
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



The distribution of highway MPG shows that the majority of vehicles fall within a common efficiency range (below 30). Only a small number of vehicles have highway MPG values above 30, which leads to a right-skewed distribution.

Exercise 2 Create a scatterplot. Put hwy on the x-axis and cty on the y-axis. Describe what you notice. Is there a relationship between hwy and cty? What does this mean?

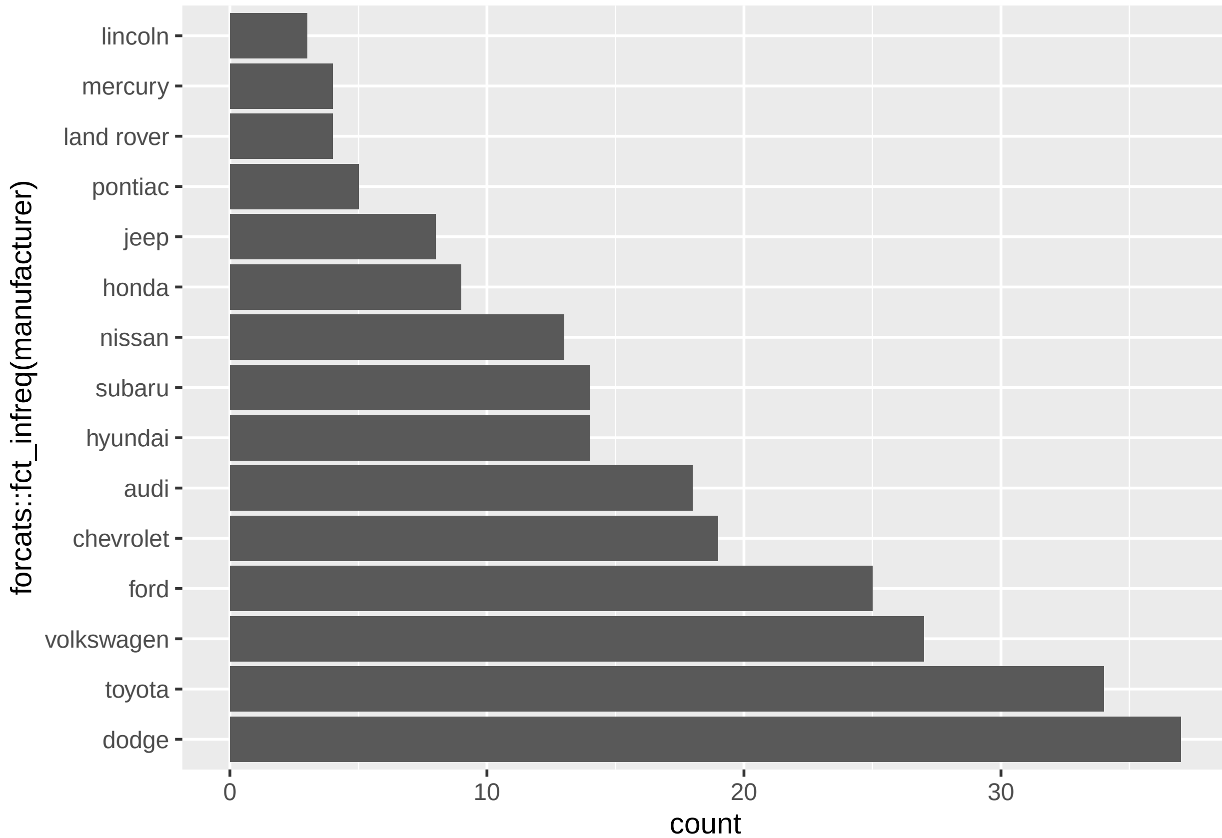
```
mpg %>% ggplot(aes(x = hwy, y = cty)) +
  geom_point()+
  geom_jitter()
```



There is a clear positive relationship between hwy and cty, which means that cars with higher highway MPG also have higher city MPG. Also, the plot shows an approximate linear trend between two variables.

Exercise 3 Make a bar plot of manufacturer. Flip it so that the manufacturers are on the y-axis. Order the bars by height. Which manufacturer produced the most cars? Which produced the least?

```
mpg %>% ggplot(aes(x = forcats::fct_infreq(manufacturer))) +
  geom_bar() +
  coord_flip()
```

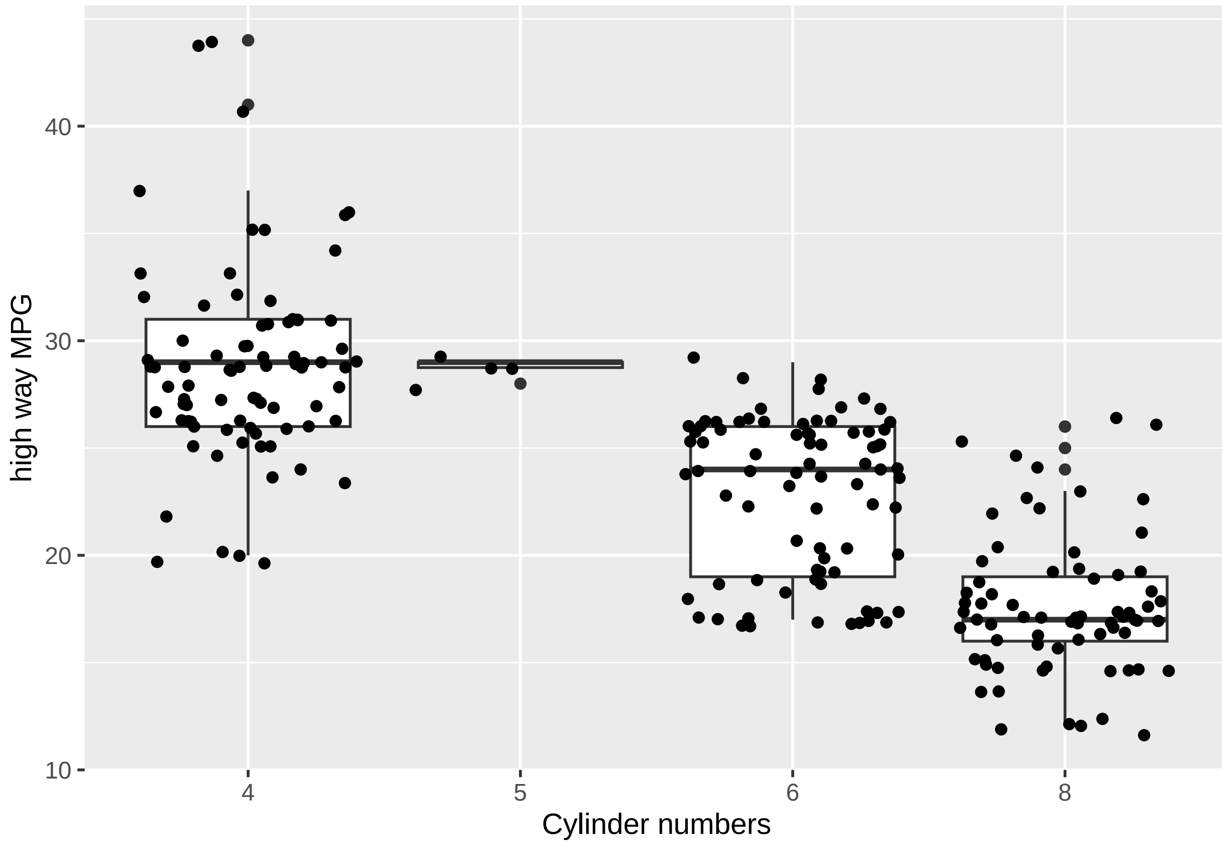


The bar plot ranks manufacturers by the number of cars they produced. From the plot, we can clearly see that Dodge produced the most cars, and Lincoln produced the fewest.

Exercise 4 Make a box plot of hwy, grouped by cyl. Use geom_jitter() and the alpha argument to add points to the plot.

Describe what you see. Is there a relationship between hwy and cyl? What do you notice?

```
mpg %>% ggplot(aes(x = factor(cyl), y = hwy)) +
  geom_boxplot() +
  xlab("Cylinder numbers") +
  ylab("high way MPG") +
  geom_jitter()
```



The plot shows a negative relationship between hwy and cyl. When the number of cylinders increases, the highway MPG decreases. Cars with fewer cylinders usually have higher highway fuel efficiency. Also, cars with 4 cylinders show greater variability in high way MPG; in contrast, cars with 5 cylinders has the fewest observations.

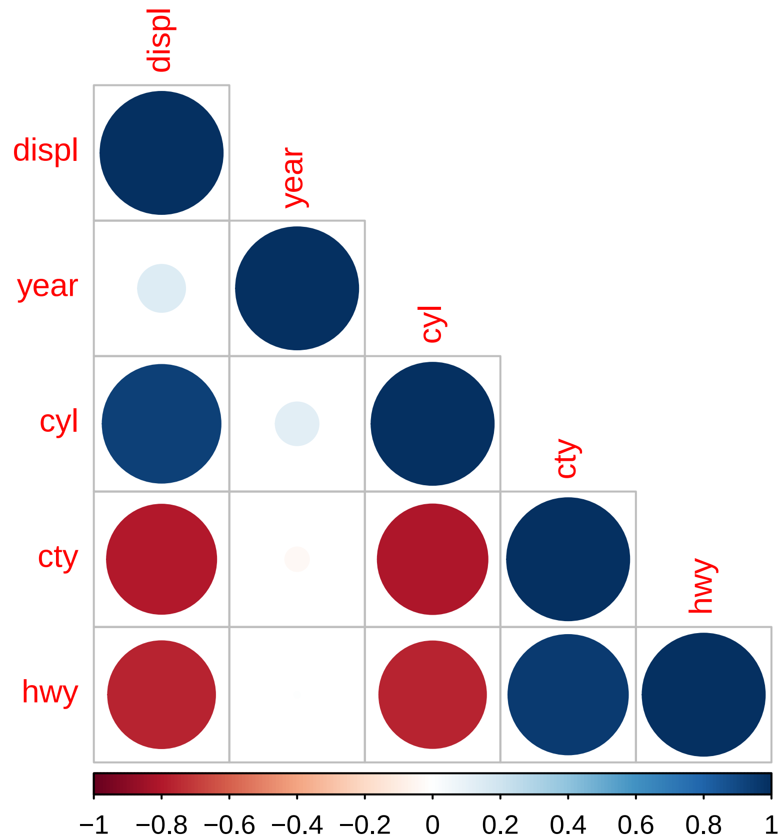
Exercise 5 Use the corrplot package to make a lower triangle correlation matrix of the mpg dataset. (Hint: You can find information on the package here.)

Which variables are positively or negatively correlated with which others? Do these relationships make sense to you? Are there any that surprise you?

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
mpg %>% select(where(is.numeric)) %>%
  cor(use = "complete.obs") %>%
  corrplot(type = "lower")
```



City MPG (cty) and highway MPG (hwy) are correlated strongly and positively. It makes sense to me and does not surprise me because both of them measure the fuel efficiency. Moreover, "year" shows a relatively weak correlations with other variables, which means that year does not have a strong relationship with fuel efficiency and engines. Also, displacement (displ) and number of cylinders (cyl) are negatively correlated with cty and hwy, meaning that vehicles with larger engines may have lower fuel efficiency.

Therefore, these relationships make sense to me, and do not surprise me.